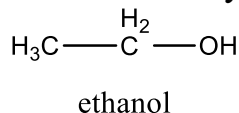


Alcohol

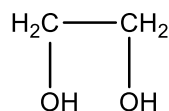
Alcohol:

The compound having OH group attached to the alkyl group area called alcohols.

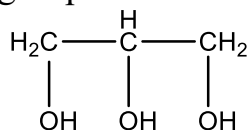


They may be monohydric having one OH group.

Polyhydric having more than one OH group.



ethane-1,2-diol

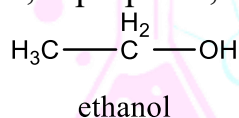


propane-1,2,3-triol

Classification of Monohydric Alcohols

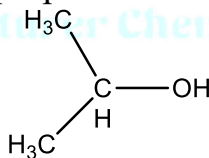
(a) Primary alcohols

In primary alcohols the carbon atom of hydroxyl group is attached to only one alkyl group. For example, methanol, ethanol, 1-propanol, etc.



(b) Secondary alcohols

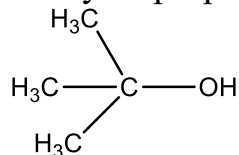
In secondary alcohols the carbon atom of hydroxyl group is attached to two alkyl groups. For example, Iso-propyl alcohol or 2-propanol.



propan-2-ol

(c) Tertiary alcohols

In tertiary alcohols the carbon atom of hydroxyl group is attached to three alkyl groups. For example, Ter-butyl alcohol or 2-Methyl-2-propanol.

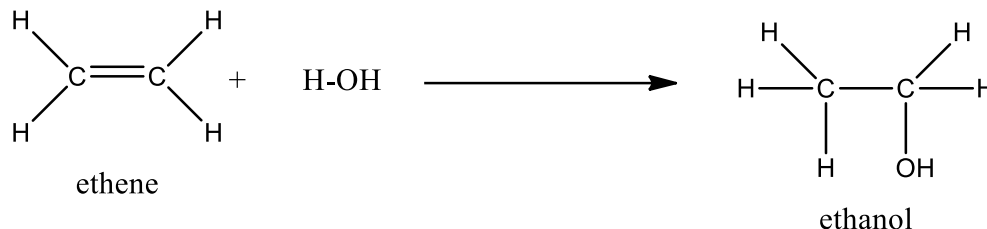


2-methylpropan-2-ol

Preparation of Alcohols

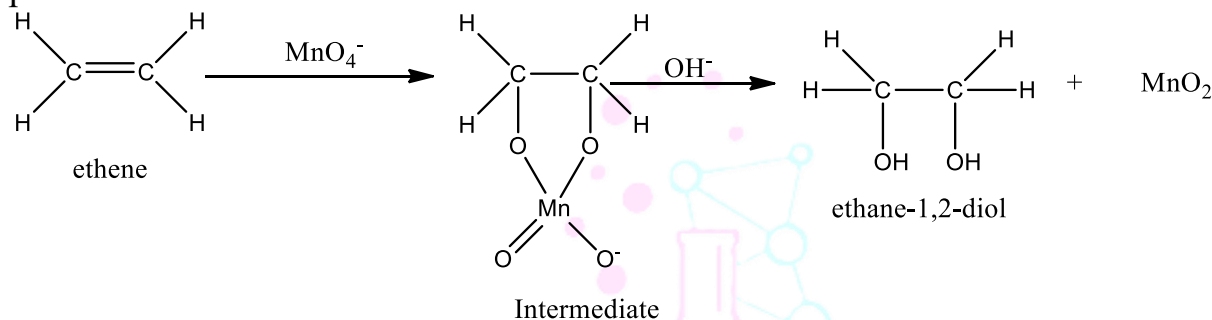
(a) By electrophilic addition of steam to an alkenes

Alkenes react with steam in presence of phosphoric acid as catalyst at 300 °C and 60-70 atm. pressure to form alcohol.



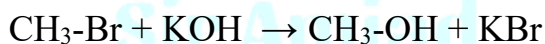
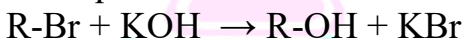
(b) By the reaction of alkenes with potassium manganate (VII)

When an alkene is treated with cold dilute acidified solution of KMnO_4 , a diol is produced.



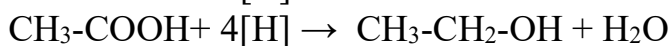
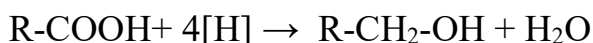
(c) By the substitution of haloalkanes

When a haloalkane is heated with aqueous solution of NaOH an alcohol is produced.



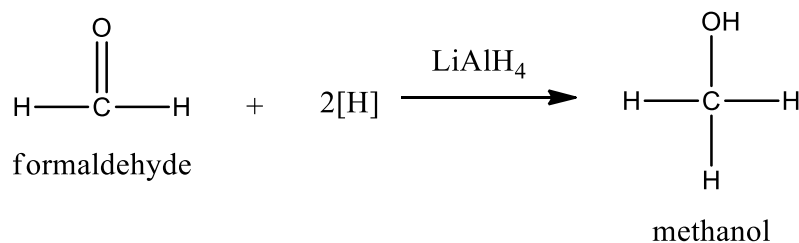
(d) By the reduction of carboxylic acids

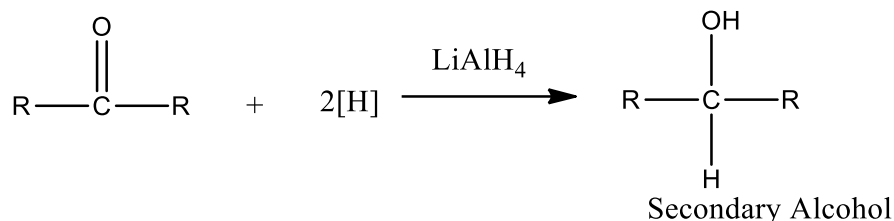
Carboxylic acids and esters can be reduced to alcohols with the help of reducing agents such as NaBH_4 or LiAlH_4



(e) By the reduction of carbonyl compounds

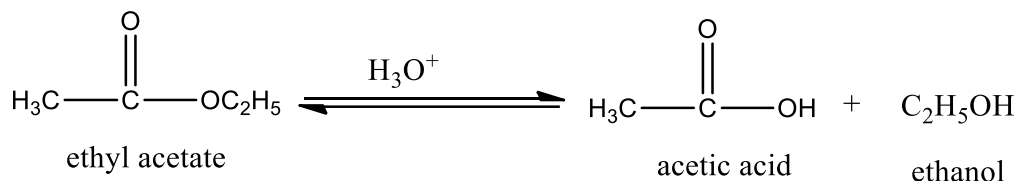
Carbonyl compounds are reduced to respective alcohols by using reducing agents such as NaBH_4 or LiAlH_4 .



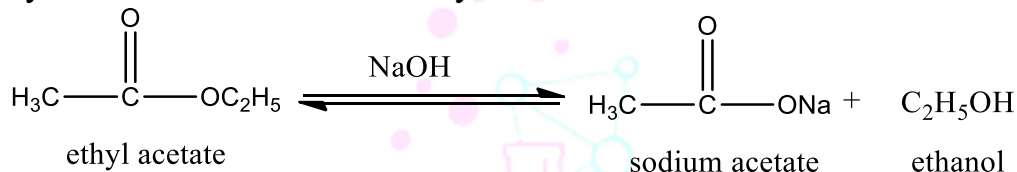
**(f) By the hydrolysis of an Esters**

Ester hydrolyze when heated with dilute acid or dilute alkali to form an alcohol.

In acid hydrolysis, the H of water joins with the OR part of ester and OH joins with carbonyl carbon atom.

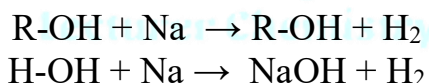


Basic hydrolysis of an ester forms carboxylate salt and an alcohol.

**Acidity of Alcohol**

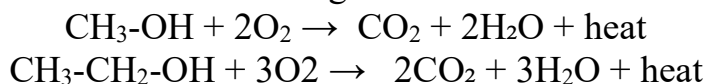
Alcohols and water both contain -OH group. The -OH group in alcohols is less polar than -OH group in water. The electron releasing nature of alkyl group decreases the polarity of -OH bond. This makes alcohols weaker acid than water.

Like water alcohols also release hydrogen with metals like sodium or potassium and form alkoxide.

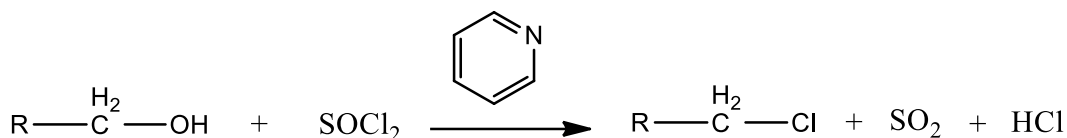
**REACTIONS OF ALCOHOLS****(a) Combustion**

Alcohols are highly inflammable compounds. They burn completely in air to form carbon dioxide and water.

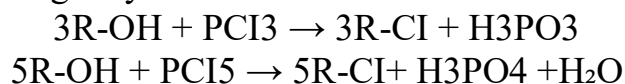
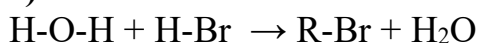
They are used as high octane fuels for racing cars.

**(b) Reaction with SOCl₂ to give Alkyl Halides**

When alcohols react with thionyl chloride in the presence of pyridine as a solvent, it gives alkyl chlorides.

**Reaction with PX₃ to give Alkyl Halides**

Alcohols react with phosphorus trichloride and phosphorous pent chloride (PCl₃ and PCl₅) to form corresponding alkyl halides.

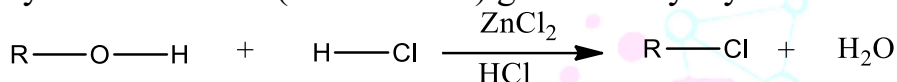
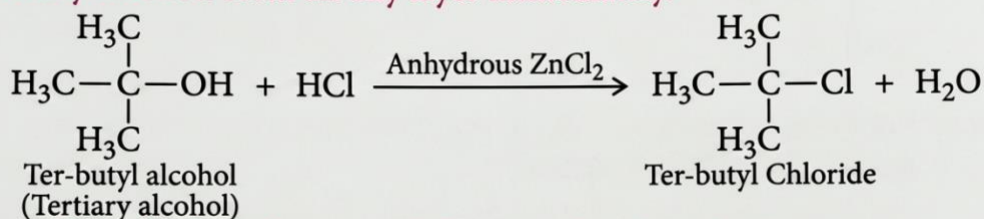
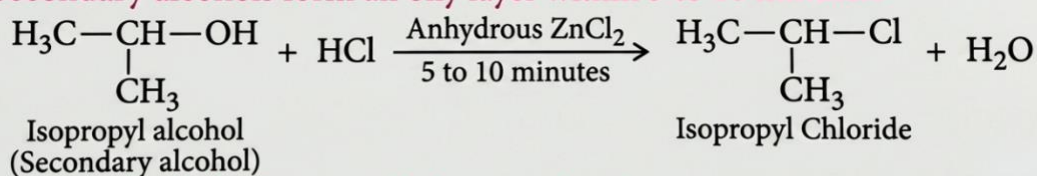
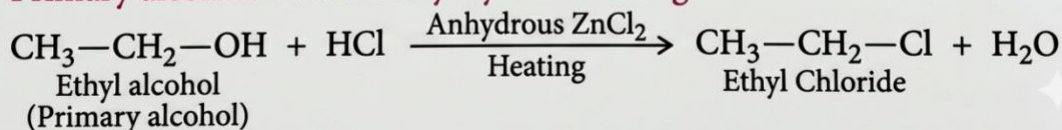
**c) Reaction with HX**

In this substitution reaction, the OH group of alcohol is replaced by a halogen atom of hydrogen halide to produce alkyl halide.

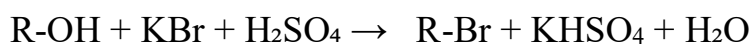
Lucas Test:

(Test used to distinguish primary, secondary and tertiary alcohols)

Alcohols on reaction with Lucas reagent anhydrous zinc chloride and concentrated hydrochloric acid (ZnCl₂ / HCl) gives an oily layer of haloalkanes.

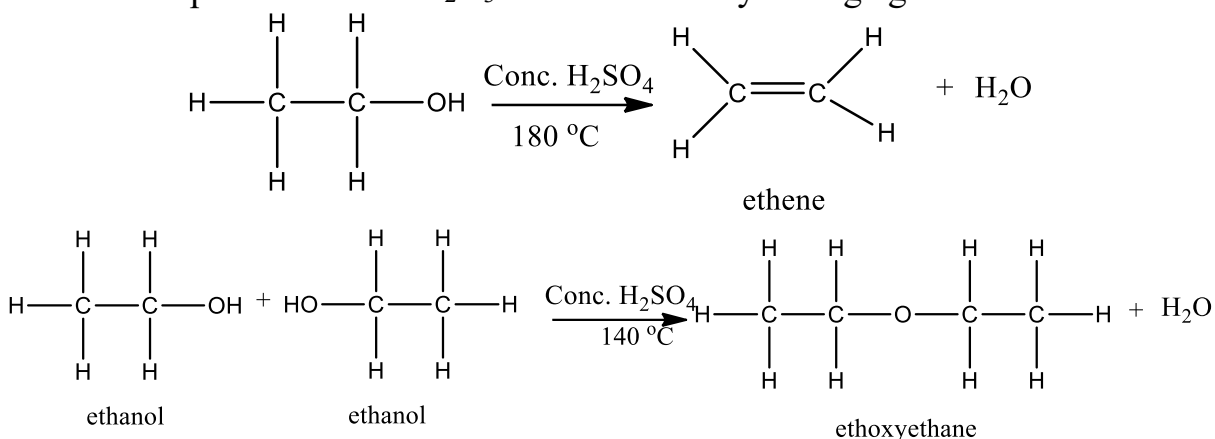
**(a) Tertiary alcohols form an oily layer immediately.****(b) Secondary alcohols form an oily layer within 5 to 10 minutes.****(c) Primary alcohols form an oily layer on heating.****(d) Reaction with potassium bromide**

When an alcohol is heated with potassium bromide and concentrated sulphuric acid forms a bromoalkane.

**(e) Dehydration of alcohol**

When dehydration of alcohol takes place, two different types of products are formed under different set of conditions.

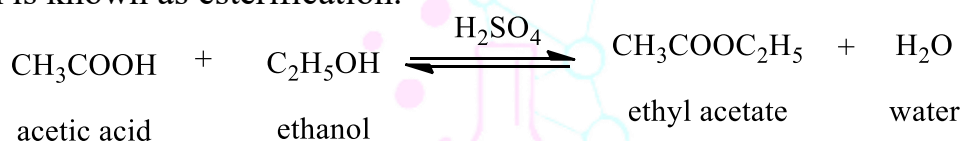
Concentrated Sulphuric acid or Al_2O_3 is used as a dehydrating agent.



(f) Formation of Esters (Reaction with Carboxylic Acid)

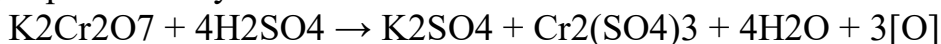
When alcohols react with carboxylic acid in the presence of concentrated sulphuric acid or Al_2O_3 as a catalyst, they give ester and water.

This reaction is known as esterification.



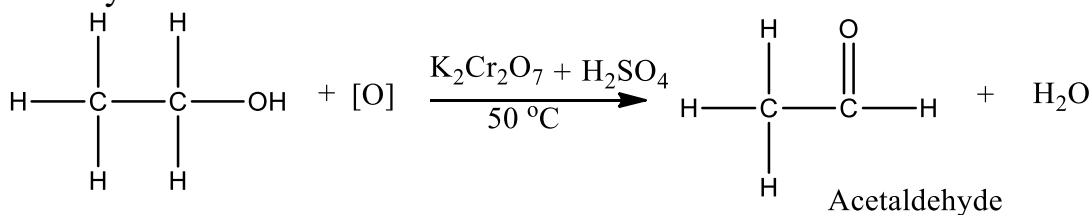
(g) Oxidation of alcohols.

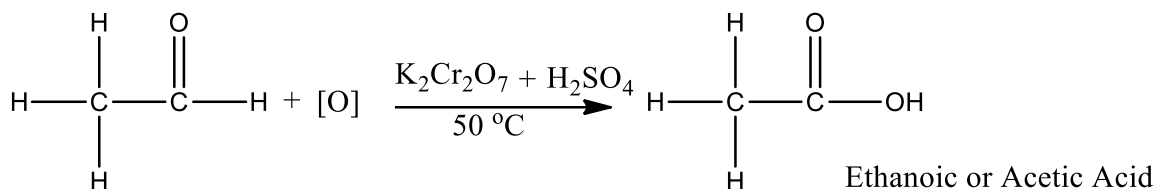
- Alcohols are oxidized by heating with a mild oxidizing agent such as potassium dichromate mixed with dilute sulfuric acid.
- During alcohol oxidation, dichromate ions $\text{Cr}_2\text{O}_7^{2-}$ (orange in colour) are reduced to chromium ions Cr^{3+} (green in colour).
- Therefore, this oxidation-reduction reaction can be easily detected.
- Nascent oxygen produced by $\text{K}_2\text{Cr}_2\text{O}_7$ and sulfuric acid oxidizes alcohols.



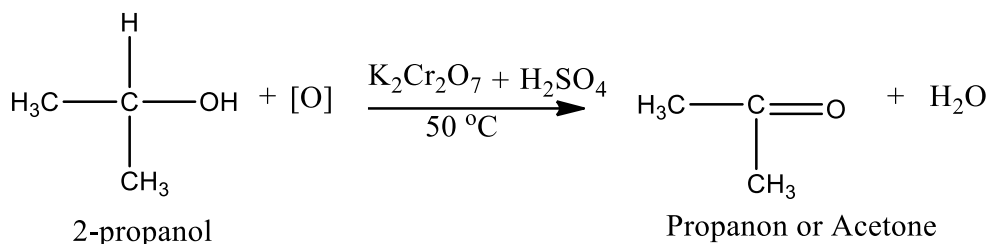
(a) Primary Alcohol:

A primary alcohol is first oxidized to an aldehyde, which is further oxidized to a carboxylic acid.

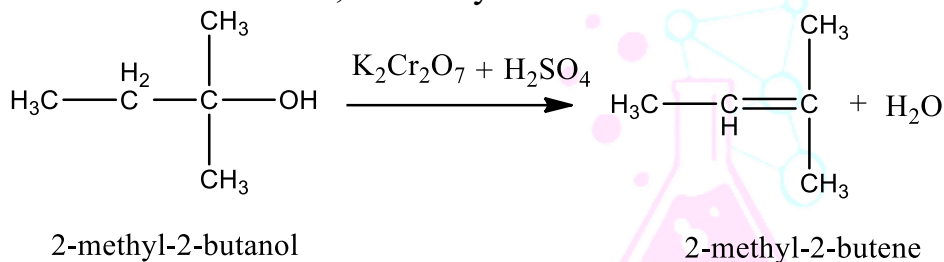


**(b) Secondary Alcohol:**

A secondary alcohol is oxidized under similar condition to give a ketone which is not further oxidized.

**(c) Tertiary Alcohol:**

A tertiary alcohol is not oxidized by acidified KMnO_4 . When heated with a mixture of $\text{K}_2\text{Cr}_2\text{O}_7$ and H_2SO_4 , it is dehydrated to form an alkene.

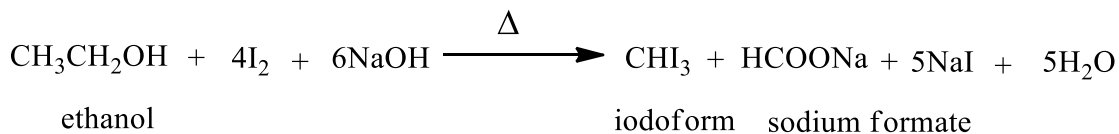
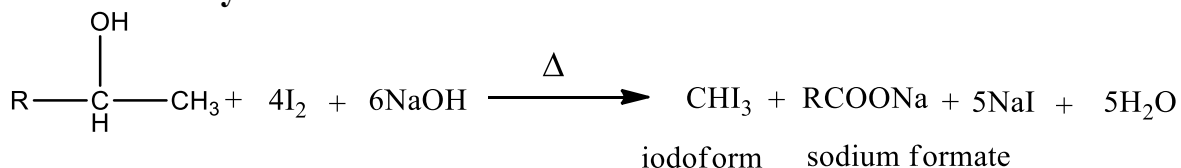
**(h) Iodoform reaction**

Iodoform reaction is characteristic reaction of ethanol and secondary alcohols.

When these alcohols are treated with a mixture of excess of iodine and excess of sodium hydroxide solution, a pale yellow precipitate of tri-iodomethane is formed. This reaction is called iodoform reaction.

Ethanal and methyl ketones also give this test.

In primary alcohol only ethanol will give this test positive.

For Ethanol**For Secondary Alcohol**

This test is not given by tertiary alcohols.

For More Information:**Online and physical tuition is available****Muhammad Amjad**WhatsApp: 0306-7395469

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